College of Industrial Technology

King Mongkut's University of Technology North Bangkok

เลขที่นั่งสอบ

Midterm Examination of Semester 1 **Year:** 2013

Subject: 341151 Electric Circuits I **Section:** 5 - 6

Date: 3 August 2013 **Time:** 8.00 - 10.00

Name:______________________ ID:____________________ Class:___________

Instructions:

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken out of the examination room.
3. Text books and dictionaries are NOT allowed.
4. Calculator is allowed.
5. Electronic commutations devices are NOT allowed in the examination room.
6. The examination has 12 pages (including this page), 3 parts (58 questions) and a total score of 70 points.
7. Write all your answers in this examination sheet.

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Part 1 (15 points) True/False

Instruction: Mark the correct answer for following questions in the provided answer sheet.

1. Coulomb's law is about force between charges.
2. Coulomb is the unit of electric charge in SI unit
3. The SI unit of inductance is named after Michael Faraday.
4. The notation ms is an abbreviation for micro seconds.
5. From English standard, MKS represents Meters, Kilometers, and Seconds.
6. For simple atom structure, nucleus contains protons and electrons.
7. Secondary cell is the battery that cannot be recharged.
8. Gold is a better conductor than copper.
9. The unit of conductance is in ohm.
10. The shorter electric wire length, the lower resistance.
11. For four-band resistor's color code, the third band represents power-of-ten multiplier.
12. Power in watt is defined by Joule per second.
13. Efficiency can be determined by input power over output power.
14. Kilowatt-hour meter is used to measure the energy of electricity supplied to the residential user.
15. The current that flows through series elements of a circuit is the same in each element.

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Part 2 (40 points) Multiple choices

Instruction: Mark the correct answer for following questions in the provided answer sheet.

1. Evaluate the following expression: $(1 \times 10^{-6}) / (1 \times 10^{-9}) =$

   (a) 1×10^{-3} (b) 1×10^{3}

   (c) 0.01 (d) 100

2. Express the number 0.0000001 using the powers of ten.

   (a) 1×10^{-5} (b) 1×10^{-7}

   (c) 1×10^{-4} (d) 1×10^{-6}

3. Four 12-volt batteries connected together by a conductor, positive terminal to negative terminal will result in which of the following voltage?

   (a) 12 volts (b) 24 volts

   (c) 36 volts (d) 48 volts

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4. In double-subscript notation, the voltage V_{ab} is the same as

   (a) $V_a - V_b$ (b) $V_a + V_b$

   (c) $V_a \times V_b$ (d) V_a

5. The voltage divider rule states that for two series ______________ of equal value, the voltage will divide equally.

   (a) elements (b) resistors

   (c) circuits (d) branches

6. An ammeter is always placed _____________ an element to measure the current.

   (a) in parallel with (b) after

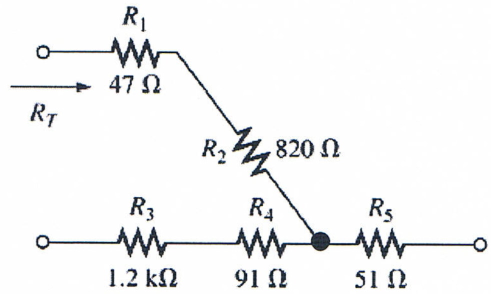
   (c) in series with (d) on top of

7. Which of the following is the list of conductors from higher conductivity to lower conductivity?

   (a) Silver – Copper – Gold – Aluminum (b) Gold – Silver – Copper – Aluminum

   (c) Silver – Gold – Aluminum – Copper (d) Gold – Silver – Aluminum – Copper

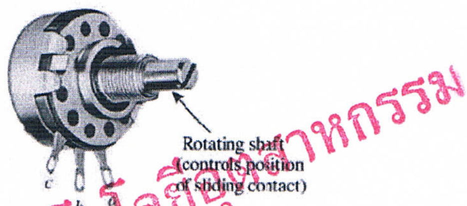
8. From Figure below, the total resistance R_T is



(a) 47 Ω + 820 Ω + 51 Ω (b) 51 Ω + 91 Ω + 1.2k Ω

(c) 47 Ω + 820 Ω + 91 Ω + 1.2k Ω (d) 47 Ω+820 Ω+51 Ω+91 Ω+1.2k Ω

9. From figure below, which terminal pair will not affect the change in resistance if use ohm meter to measure.



(a) a and b (b) b and c (c) a and c (d) none

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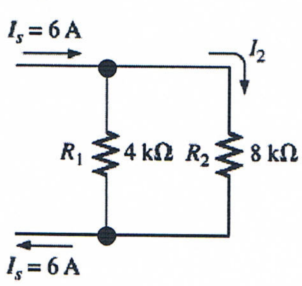
10. What is the symbol of photoresistor?

(a)  (b)  (c)  (d) 

11. From Figure below, the value of I_2 is

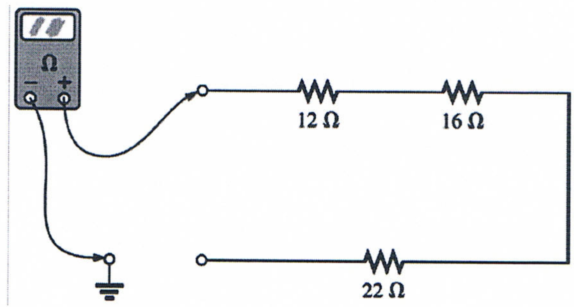


(a) 0 A (b) 2 A (c) 4 A (d) 6 A

12. What is the power deliver to a resistor 10 kΩ with voltage across of 10 volts?

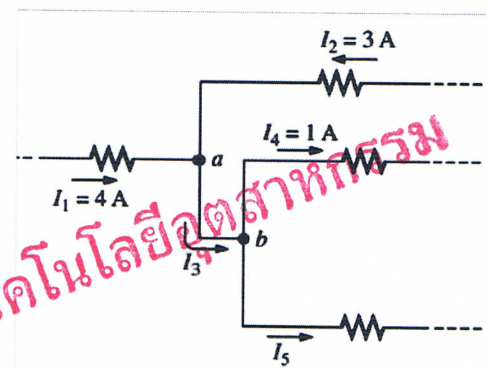
(a) 1 mW (b) 10 mW (c) 100 mW (d) 1 W

13. From figure below, what is the ohmmeter reading?



(a) 0 ohms | (b) 50 ohms
(c) 12 ohms | (d) Infinity ohms

14. The value of I_3 is __________ and I_5 is __________.



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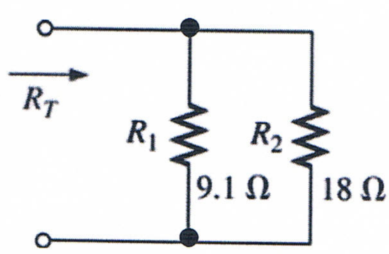
(a) 7 A and 6 A | (b) 7 A and – 6 A
(c) – 7 A and 6 A | (d) – 7 A and – 6 A

15. From figure below, what R_T should be?

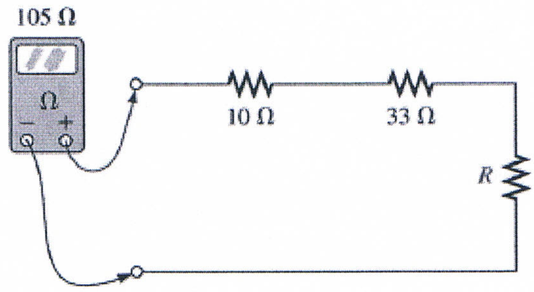


(a) Lower than 9.1 ohms. | (b) Slightly lower than 18 ohms.
(c) Higher than 18 ohms. | (d) Lower than 18 ohms but slightly higher than 9.1 Ω.

16. How much charge passes through a battery of 6.3 V if the energy expended is 80 J?

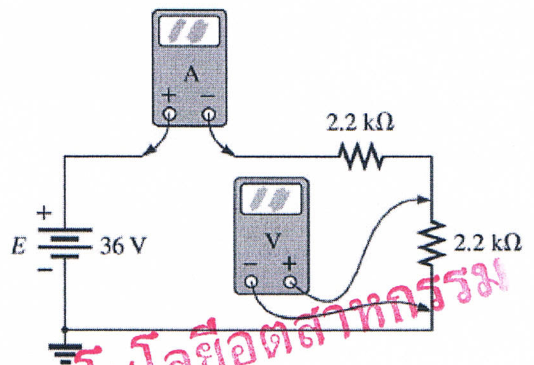
(a) 13.5 C (b) 12.7 C (c) 10.6 C (d) 11.4 C

17. From Figure below, the value of R is



(a) 43 Ω (b) 62 Ω (c) 82 Ω (d) 148 Ω

18. From Figure below, which of the following gives the value of voltmeter reading?



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(a) 12V (b) 18V (c) 36V (d) 48V

19. From previous problem, which of the following gives the value of ammeter reading?

(a) 0 mA (b) 2.2 mA (c) 8.18 mA (d) 16.36 mA

20. Potential is defined as the __________ at a point with respect to another point in the electrical system.

(a) charge (b) current

(c) ampere (d) voltage

21. Which of the following expresses a double-superscript notation of V_{ab} =-30 V?

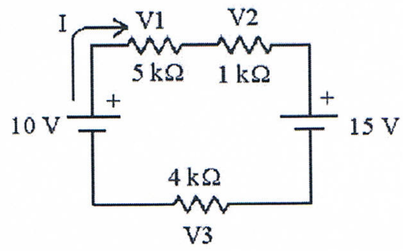
(a) Point a has a higher potential than b.

(b) Point a has the same potential as b.

(c) Point a has a lower potential than b.

(d) Point a cannot be determined.

22. See Figure below, which one of the following KVL equations describes this circuit?



(a) +10 V + V1 + V2 - 15 V + V3 =0

(b) +10 V - V1 - V2 + 15 V - V3 = 0

(c) +10 V - V1 - V2 - 15 V - V3 = 0

(d) -10 V - V1 - V2 + 15 V - V3 =0

23. From previous problem, the total current I is

(a) -500 μA
(b) +500 μA
(c) -2.5 mA
(d) +2.5 mA

24. A circuit has a source voltage of 15 volts and two resistors with a total resistance of 4000 ohms. If R_1 has a potential of 9.375 volts, what is the value of R_2?

(a) 500 ohms
(b) 1000 ohms
(c) 1500 ohms
(d) 2000 ohms

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25. Insulators are materials that have ____________ free electrons and require a large applied potential to establish a measurable current level.

(a) several
(b) very few
(c) many
(d) some

26. The unit of the "horsepower" is related to the SI unit of power as 1 horsepower = _____________ watts.

(a) 674
(b) 476
(c) 647
(d) 746

27. Parallel elements ___________ without changing the total resistance or input current.

(a) cannot be interchanged
(b) cannot be moved
(c) may be removed
(d) can be interchanged

28. For how many hours will a battery with an Ah rating of 32 theoretically provide a current of 1.75 A?

(a) 18.3 h (b) 17.3 h (c) 19.3 h (d) 20.3 h

29. If a resistor uses 420 J of energy for 7 min, what is the power to the resistor?

(a) 1 W (b) 80 mW (c) 80 W (d) 1 mW

30. A 250 W light bulb operates on 120 V. How much current is flowing through the bulb?

(a) 20.8 S (b) 2.08 mA (c) 0.208 S (d) 2.08 A

31. What is the cost of using a 24 W radio for 1 hour at a rate of 2 bahts per kilowatthour?

(a) 0.024 baht (b) 0.24 baht (c) 0.048 baht (d) 0.48 baht

32. Find the total conductance for the network.

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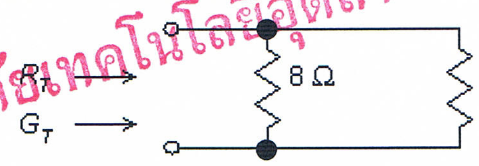
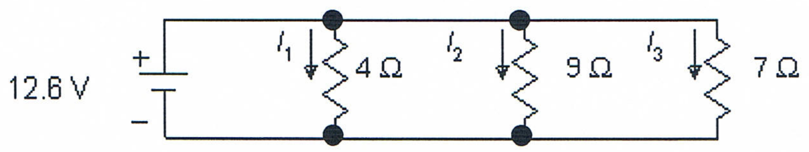


(a) 187.5 mS (b) 187.5 S (c) 18.75 mS (d) 18.75 S

33. For parallel elements, the total conductance is the ______ of the individual conductance.

(a) sum (b) parallel total (c) algebraic sum (d) series total

34. Find the current I_3.

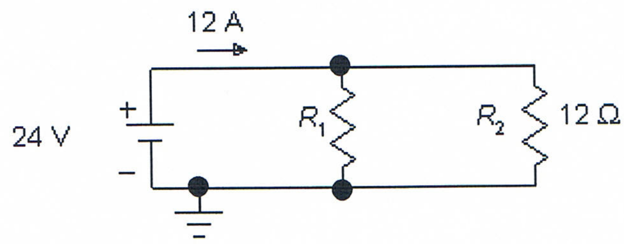


(a) 18 A (b) 1.8 A (c) 18 mA (d) 0.18 mA

35. A wattmeter is an instrument used to measure ________________.

(a) power (b) current (c) resistance (d) voltage

36. Determine the resistance R_1.



(a) 0.24 Ω (b) 240 Ω (c) 24 Ω (d) 2.4 Ω

37. The current divider rule is similar to the voltage divider rule except that the current ____________ the resistances.

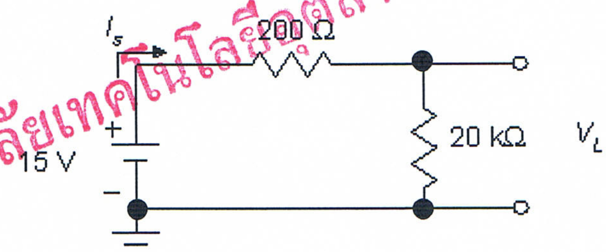
(a) divides in an inverse relationship to the values of

(b) divides in a direct relationship to the values of

(c) always divides equally among

(d) divides independent of the values of

38. Determine I_S.



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(a) 743 mA (b) 0.743 A (c) 0.743 mA (d) 7.43 mA

39. From previous problem, determine V_L if the 20 kΩ resistor is shorted out.

(a) 7.5 V (b) 0 V (c) −7.5 V (d) 15 V

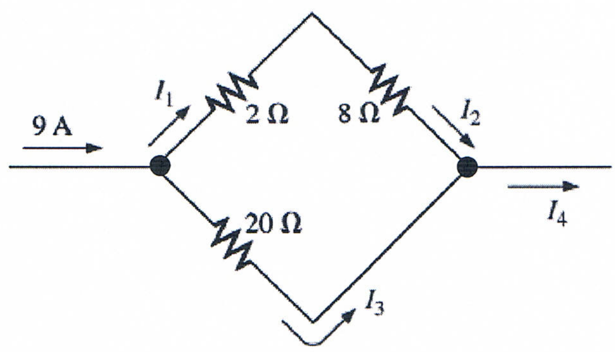
40. What is the efficiency of a motor that has an output of 1.5 hp with an input of 1550 W?

(a) 0.722 (b) 7.22 (c) 0.139 (d) 1.39

Part 3 (15 points) Calculation

Instruction: Show the mathematical expression and answer the following problems.

1. From Figure below, determine the value current I_1, I_2, I_3, and I_4



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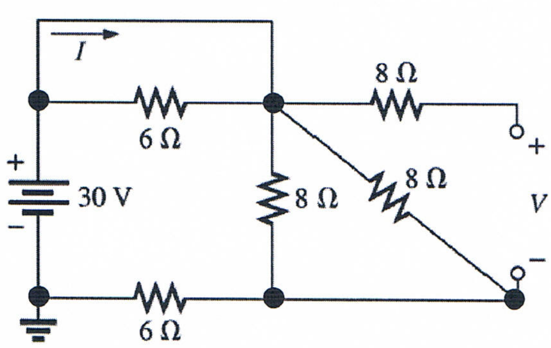
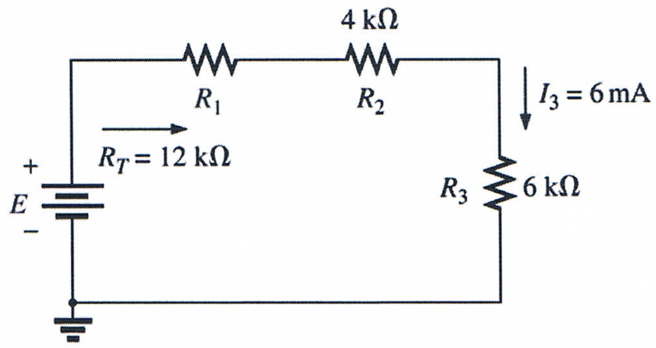
2. From Figure below, find the voltage V and current I.



3. Find the value of E and R_1



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Answer Sheet

Name:____ID:__________________

Subject: 341151 Electric Circuits I

Part 1

	True	False
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Part 2

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